

Artificial Intelligence in Pediatric Radiology

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CHLA Radiology Grand Rounds

October 8, 2026

Objectives

- Describe the big players in radiology AI, both general and pediatric
- Summarize what radiology AI does well and what remains unresolved

Methods

Journals, Preprints: PubMed, Embase, arXiv, medRxiv

Conferences: NeurIPS, ICLR, ICML, CVPR, MICCAI, MIDL

Datasets: Hand-curated list

Newsletters: The Imaging Wire, RSNA News, TLDR AI, TLDR Tech, Signify Research, ESR / ECR, Radiology Business

Commercial: FDA list of AI-enabled devices

Date range: January 1 2008 – September 30 2026

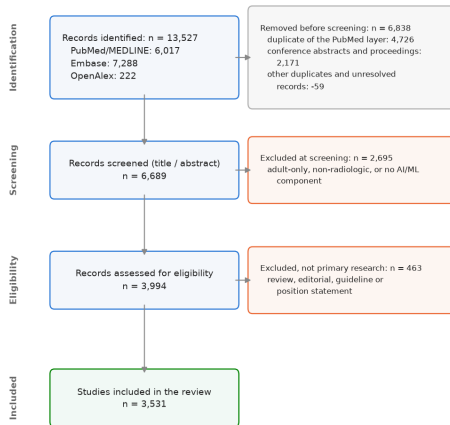
Search strategy: radiology AND AI [AND pediatric]

radiology: (radiology[tiab] OR radiological[tiab] OR radiograph*[tiab] OR "medical imaging"[tiab] OR "diagnostic imaging"[tiab] OR tomography[tiab] OR "magnetic resonance"[tiab] OR MRI[tiab] OR "MR imaging"[tiab] OR "MR images"[tiab] OR MRIs[tiab] OR "MR image"[tiab] OR "MR scan"[tiab] OR "MR scans"[tiab] OR "computed tomography"[tiab] OR CT[tiab] OR ultrasound[tiab] OR ultrasonograph*[tiab] OR sonograph*[tiab] OR echocardiograph*[tiab] OR mammograph*[tiab] OR "chest x-ray"[tiab] OR "x-ray"[tiab] OR fluoroscop*[tiab] OR scintigraph*[tiab] OR "positron emission"[tiab] OR PET[tiab] OR SPECT[tiab] OR angiograph*[tiab] OR neuroimaging[tiab] OR tractograph*[tiab] OR elastograph*[tiab] OR "fMRI"[tiab] OR "functional MRI"[tiab] OR "functional magnetic resonance"[tiab] OR connectome*[tiab] OR "diffusion tensor"[tiab] OR DTI[tiab] OR "bone age"[tiab] OR "skeletal age"[tiab] OR "skeletal maturity"[tiab] OR "barium enema"[tiab])

AI: ("artificial intelligence"[tiab] OR "machine learning"[tiab] OR "deep learning"[tiab] OR "convolutional neural network"[tiab] OR "convolutional neural networks"[tiab] OR "neural network"[tiab] OR "neural networks"[tiab] OR "computer-aided diagnosis"[tiab] OR "computer aided diagnosis"[tiab] OR "computer-aided detection"[tiab] OR radiomic*[tiab] OR "computer vision"[tiab] OR "large language model"[tiab] OR "large language models"[tiab] OR "foundation model"[tiab] OR "foundation models"[tiab] OR "transfer learning"[tiab] OR "vision transformer"[tiab] OR "self-supervised"[tiab] OR "random forest"[tiab] OR "support vector machine"[tiab] OR "gradient boosting"[tiab] OR "natural language processing"[tiab])

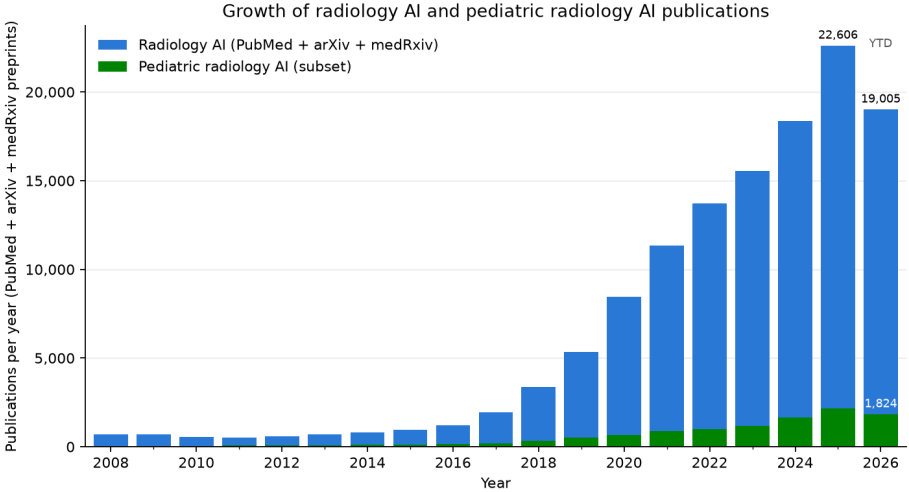
pediatric: (pediatric*[tiab] OR paediatric*[tiab] OR child*[tiab] OR infant*[tiab] OR neonat*[tiab] OR adolescen*[tiab] OR fetal[tiab] OR foetal[tiab] OR fetus*[tiab] OR foetus*[tiab] OR newborn*[tiab] OR preterm[tiab] OR "premature infant"[tiab] OR "premature infants"[tiab] OR "premature birth"[tiab] OR "premature neonate"[tiab] OR "premature neonates"[tiab] OR prenatal[tiab] OR antenatal[tiab] OR perinatal[tiab] OR youth[tiab] OR juvenile[tiab] OR "children's hospital"[tiab] OR schoolchild*[tiab] OR toddler*[tiab] OR baby[tiab] OR babies[tiab])

Study selection

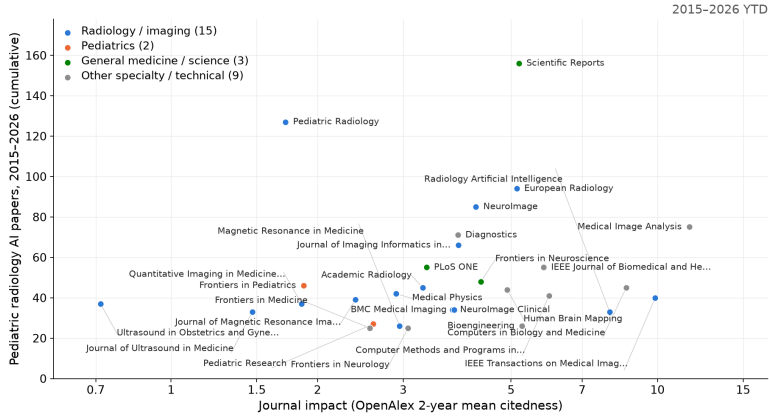


PubMed and Embase layers complete (2005-2026). Conference material is retrieved and reported but excluded by the publication-form criterion.

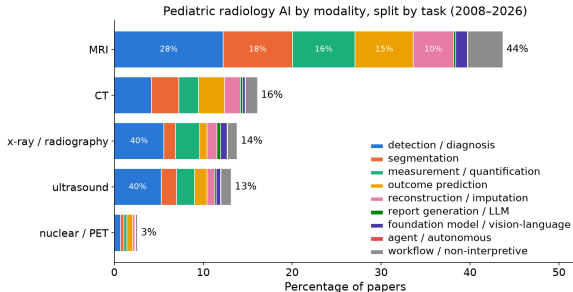
Radiology AI journal publication counts



Which journals publish pediatric radiology AI



Pediatric radiology AI publications, stratified by modality and task

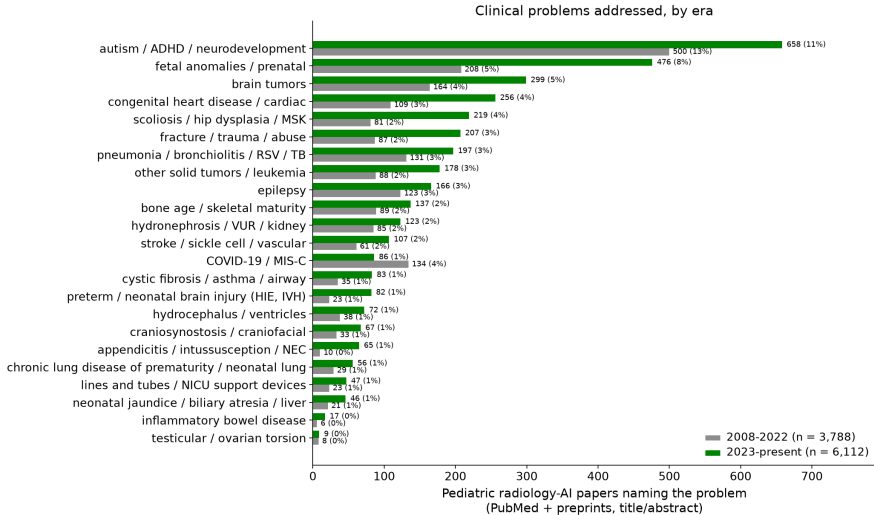


- **detection / diagnosis:** predict discrete disease labels (e.g., fracture, pneumonia, tumor, triage)
- **segmentation:** outline a structure or lesion
- **measurement / quantification:** measure a number from the image (e.g., bone age, Cobb angle, organ volume, fetal biometry)

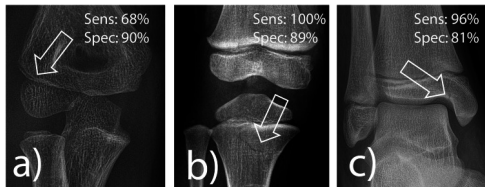
- **outcome prediction:** a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- **reconstruction / imputation:** generate a better or missing image
- **report generation / LLM:** text: draft, summarize, or extract from radiology reports

- **foundation model / vision-language:** a large pretrained model reused across tasks
- **agent / autonomous:** multi-step actions taken without a human in the loop
- **workflow / non-interpretive:** protocoling, scheduling, ordering, decision support, education

Pediatric radiology AI publications, stratified by medical problem



AI fracture detection validation of RBFrature (Radiobotics)

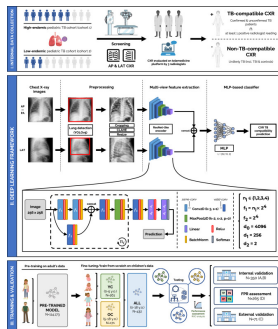


Ziegner et al., Eur Radiol 2025, figure 4: Frequently missed fracture entities with medicolegal significance according to van Laer. a Radial condyle fracture, (b) fracture of the proximal tibia, and (c) fracture of the medial malleolus.

source: doi.org/10.1007/s00330-025-11554-9

- **Problem:** pediatric fracture detection
- **Data:** 1,672 consecutive radiographs of children under 18 from one tertiary pediatric emergency department, plus a selected medicolegal set (proximal tibia, radial condyle); three pediatric residents read each before and after seeing the AI.
- **Method:** RBFrature (Radiobotics), a commercially available, CE-marked deep-learning fracture detector (adult-derived, pediatric extension) was run stand-alone and as a second reader.
- **Result:** stand-alone sensitivity 92%, radial condyle fracture sensitivity 68%, proximal tibia fracture sensitivity 100%, medial malleolus fracture sensitivity 96%. As second reader, AI raised resident sensitivity 84%→87% and accuracy 88%→90%; in 2% of cases readers dropped a correct diagnosis after seeing the AI.
- **Why it matters:** An external validation of a commercial pediatric fracture AI shows good performance, but also that even a high-performing AI can mislead readers in a small fraction of cases.

pTBLightNet: pediatric tuberculosis on frontal and lateral chest radiographs

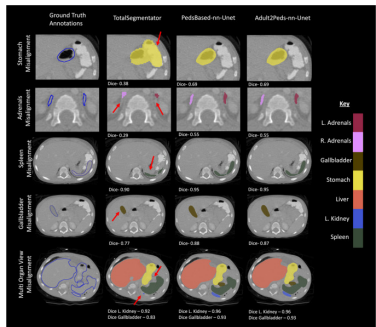


Capellan-Martin et al., Nat Commun 2025, figure 1: the multi-view pTBLightNet pipeline

source: doi.org/10.1038/s41467-025-64391-1

- **Problem:** pediatric pulmonary TB detection on chest radiographs
- **Data:** 114,173 adult chest radiographs for pre-training; 918 radiographs from three pediatric TB cohorts for fine-tuning and testing, including lateral views.
- **Method:** pTBLightNet, a deep-learning model using frontal and lateral views to detect TB-compatible findings; separate models for children under 5 and ages 5–18.
- **Result:** AUC 0.90 on internal testing and 0.68 on external testing; lateral views helped most in younger children. Age-specific models performed similarly to larger models trained across age groups.
- **Why it matters:** Adult pre-training can help with pediatric TB detection, but performance was substantially lower on external data.

Validation of adult-trained TotalSegmentator on pediatric CT

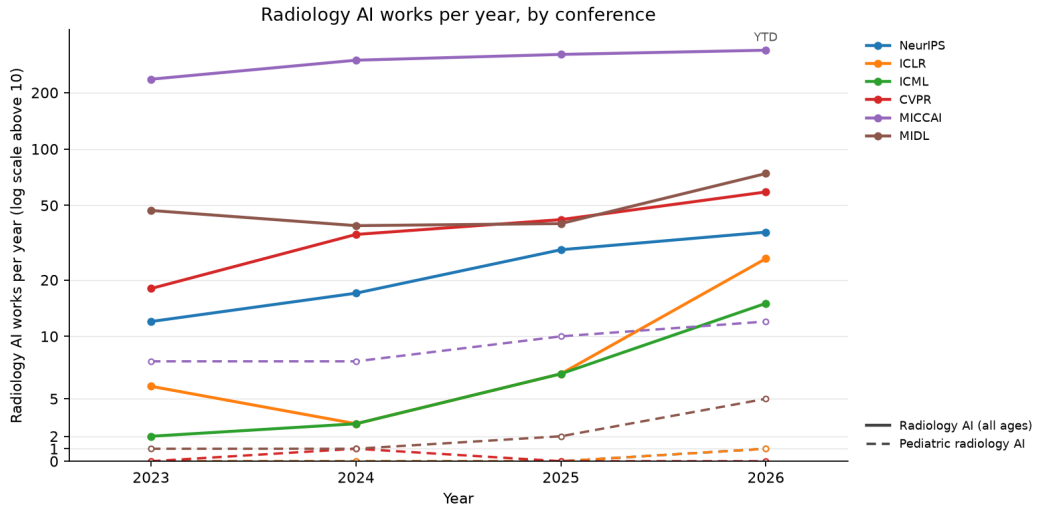


Chatterjee et al., J Imaging Inform Med 2025, figure 7: pediatric CT organ labels, ground truth vs adult-trained TotalSegmentator vs pediatric-trained models

source: doi.org/10.1007/s10278-024-01273-w

- **Problem:** abdominal organ segmentation on pediatric CT using an adult-trained model
- **Data:** 300 adult and 359 pediatric abdominal CT scans with expert organ labels for external testing.
- **Method:** TotalSegmentator, an open-source model trained on adults, was tested on both groups and compared with models trained or fine-tuned on pediatric scans.
- **Result:** Mean Dice score was 0.81 in adults and 0.73 in children, with the largest losses in the adrenals, duodenum and pancreas. Pediatric training or fine-tuning recovered the loss in performance.
- **Why it matters:** An adult-trained segmentation model performed worse in children; training on pediatric data improved performance.

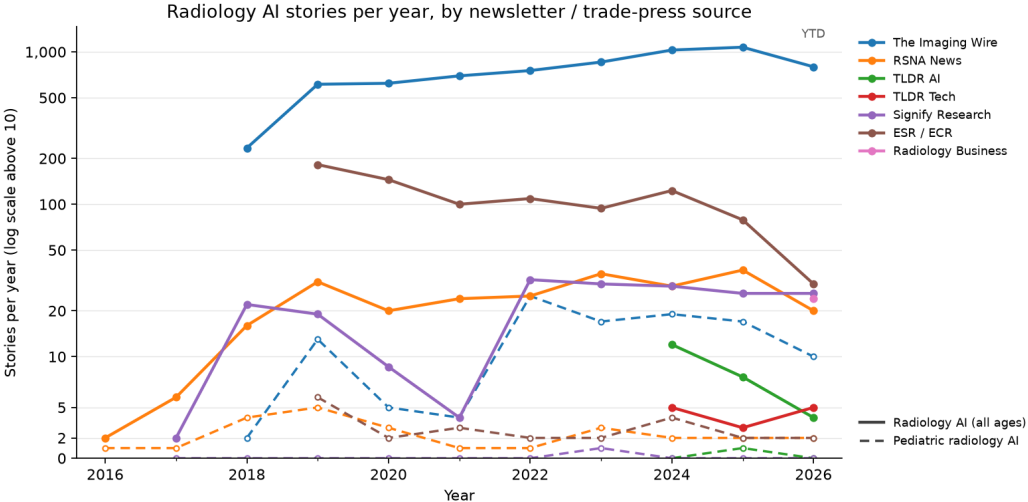
Radiology AI conference publication count



The biggest public pediatric radiology datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	< 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermany)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Radiology AI newsletter story count



Pediatric radiology AI in the trade press, 2026 (year to date) (14 stories)

Date	Newsletter	Story	Paper
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M	
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines	Prabhakaran et al., 2026 (Stroke)
2026-01-28	ESR / ECR	Guerbet: Innovation and French Cooperation in the Spotlight at ECR 2026	
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose	
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI	
2026-02-26	ESR / ECR	MRI Unveiling at the European Congress of Radiology (ECR) 2026	
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod	
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET	
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the Adult Lifespan	
2026-06-28	The Imaging Wire	BrightHeart's Bright Future	Zelop et al., 2025 (Obstetrics & Gynecology)
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging	Shah et al., 2026 (European Radiology)
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Disk	
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard	
2026-09-23	The Imaging Wire	Repeat MRI Scans Take Operational Toll	

AI use in pediatric radiology departments

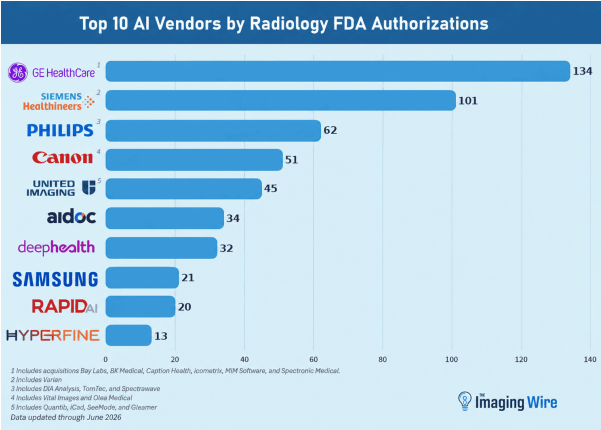
The Imaging Wire, 12 July 2026: “AI’s Rise in Pediatric Imaging” — Shah et al., *European Radiology* 2026 (doi 10.1007/s00330-026-12728-9)

- An international survey of **18 academic pediatric radiology centers** (69% response rate) asked which AI tools departments were using.
- **88.9%** reported using at least one AI tool. The most common uses were bone age (44.4%) and segmentation / quantification (22.2%). Only **16.7%** described AI as “transformational”.
- The main barriers were a lack of pediatric datasets (83.3%), PACS and workflow integration (66.7%), and cost (50%).
- Bone age is also the most common clinical application in our literature review and has an established commercial product.

Departments commonly reported using AI, but pediatric data and workflow integration remain barriers to broader use.

Radiology leads in FDA AI authorizations

The Imaging Wire, 14 September 2026: FDA AI-enabled device list, updated through June 2026



- **The FDA list:** the FDA's public list of AI-enabled devices authorized for the US market (510(k), De Novo, PMA). One row per authorization, scanners included; updated a few times a year, so it trails the newest decisions.
- **1,614** AI-enabled devices authorized since the FDA began tracking, up 5.9% from Q1 2026.
- **1,230** of them (76%) are radiology.
- Q2 2026: 89 authorizations, 66 (74%) radiology. Rules for generative AI in direct care are still pending.

Top 10 vendors by radiology FDA authorizations, counts include acquisitions

source: theimagingwire.com

European radiology AI products: the Health AI Register

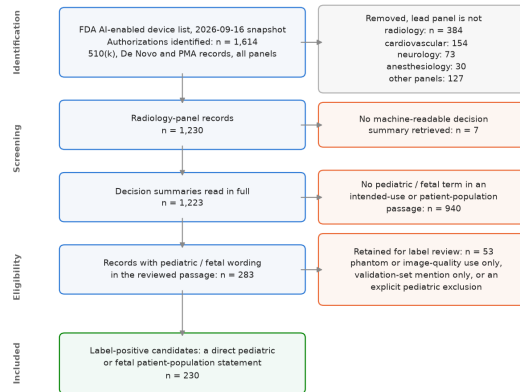
Health AI Register, radiology products (healthairegister.com), accessed 2 October 2026, matched against this deck's FDA radiology-panel list

- **The register:** an independent list of CE-marked radiology AI *software*: **309 products from 121 companies**, each with its CE class, intended use and any FDA clearance. Unlike the FDA list, it excludes scanners.
- **Overlap:** 163 of the 309 (53%) also report FDA clearance. About 135 match a specific authorization in the FDA list (35 by submission number, the rest by company and product name). The other 146 are CE-marked only.
- **FDA coverage:** companies in the register hold about 40% of the 1,230 FDA radiology authorizations. Most of the rest are scanners and US- or Asia-only vendors.
- **Recent updates:** BoneXpert was FDA-cleared on 22 September 2026 (K262390) and is not yet on the FDA list. The register still shows no US clearance for five products that the FDA list has (e.g., Gleamer BoneMetrics, Viz ANEURYSM).
- **Pediatric:** four of the 35 number-matched products carry a pediatric label in the FDA list (AZtrauma, MediAI-BA, EchoConfidence, 1CMR PRO). Several pediatric tools are CE-marked only: Gleamer BoneAge, AZboneage, Milvue TechCare Kids, Med-BoneAge.

Check both lists when reviewing a European product; either may be missing recent authorizations.

How many FDA authorizations include pediatric labeling?

FDA AI-enabled device list: which authorizations name children

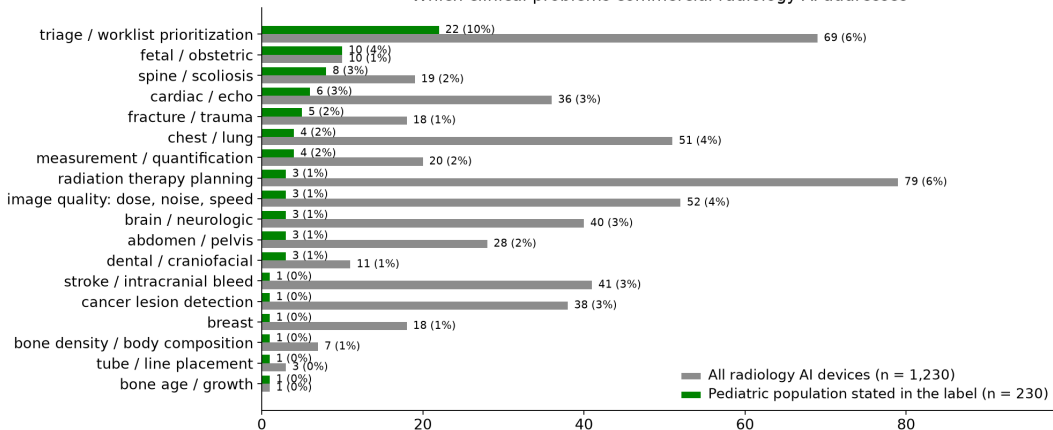


Keyword screen of the linked FDA decision summaries, not a determination of pediatric authorization: confirm the current labeling before purchase. One row per submission, so versions and resubmissions count separately.

230 of the 1,230 radiology-panel authorizations (19%) name a pediatric or fetal population in the labeling passage reviewed.

Pediatric radiology AI commercial products, stratified by medical problem

Which clinical problems commercial radiology AI addresses



FDA-authorized AI-enabled devices, Radiology panel
 (a further 689 devices, 150 of them pediatric-labeled, name no clinical problem in the device name or product code and are not shown)

Commercial products for pediatric imaging

Clinical application	Representative products	Pediatric scope / key limitation
Bone age / growth	BoneXpert; EFAI Bone Age Pro; Physis	Built for children; authorized ages and US availability vary by product.
Fracture detection	BoneView; Rayvolve; TechCare Trauma	Pediatric labeling varies by age and anatomy; BoneView and Rayvolve start at age 2.
Fetal ultrasound	Fetal EchoScan; Sonio Detect / Suspect; Delivery Date AI	Prenatal tools for different tasks: anomaly detection, view quality, or delivery prediction.
Faster MRI / lower-dose CT	GE, Siemens, Canon, Philips reconstruction; SubtleMR / SwiftMR	Pediatric protocols and evidence vary by product and version.
Spine / scoliosis	spineEOS / VEA Align; CoLumbo; Acorn 3D	Measurement and planning tools; pediatric labeling does not establish pediatric performance.
CT / CTA triage	Aidoc BriefCase; Viz LVO / ICH	Aidoc: pediatric statements, adult performance data. Viz: no pediatric statement found in reviewed summaries.
Cardiac MRI	Arterys Cardio DL	Segmentation and quantification; US labeling includes neonates through adolescents.

Selected examples. Pediatric applicability depends on the exact product, version, age, and anatomy. Full product details and FDA references are in the backup tables.

BoneXpert: automated bone age

Visiana, Denmark (bonexpert.com) — bone age / growth

- **Function:** estimates bone age from a hand radiograph using Greulich–Pyle or Tanner–Whitehouse methods. Additional features include a Bone Health Index and adult-height prediction; availability varies by market.
- **Clinical use:** developed for children and used routinely in Europe since 2009; often used as a comparator in bone-age studies.
- **US clearance:** Visiana announced FDA 510(k) clearance in September 2026 (K262390) for Greulich–Pyle bone-age assessment in patients aged **2–21 years**.
- **US availability:** release is planned for early October. The vendor states that some features, including the Bone Health Index, are unavailable in the US.

Source: Visiana clearance announcement, 30 September 2026.

Rayvolve: pediatric fracture detection

AZmed, France (azmed.co) — fracture / trauma; US pediatric fracture indication K240845 (2024)

- **What it does:** flags fractures on trauma radiographs; sibling modules add chest findings, orthopedic measurements (leg alignment, Cobb angle, hip angles) and bone age. **US pediatric clearance is fracture detection from age 2**; bone age is CE-marked only.
- **Pediatric validation:** 2,549 children in a pediatric ED (Dupuis et al., *Diagn Interv Imaging* 2022): sensitivity 95.7%, specificity 91.2%; sensitivity was **significantly lower in children under 5 and in casted limbs**.
- **Comparison with BoneView** (Robin et al., *Pediatr Radiol* 2026, 366 children): AUC 0.87 vs 0.86; Rayvolve more specific (93% vs 87%), BoneView more sensitive (85% vs 81%); both helped junior residents. BoneView's US pediatric label excludes pelvis, hip, femur, ribs and spine.
- **Reader errors:** in a pediatric ED, readers abandoned a correct diagnosis after seeing AI output in 2% of cases (Ziegner et al., *Eur Radiol* 2025).

Test fracture detection locally by age and anatomy, and decide how readers should handle disagreements with AI.

Kamran review: pediatric AI research and gaps

Kamran et al., *Pediatric Radiology* 2026 (doi 10.1007/s00247-025-06462-5)

- **Studies reviewed:** 789 pediatric-focused original articles (2005–August 2024); 91% addressed image interpretation/diagnosis. Leading applications were bone age (88), pneumonia (65), and brain lesions/tumors (54).
- **Validation:** 65% relied on local hospital datasets. The authors call for diverse, multicenter datasets and prospective validation across pediatric ages, sites, and disease-prevalence settings.
- **Research gaps:** few studies addressed communication, education, policy, or the views of children, families, and clinical teams. The authors recommend involving these groups in development and evaluation.

The review describes what has been studied. It does not compare model accuracy or determine which tools are ready for clinical use.

ACR white paper: limited access to pediatric AI

Sammer et al., *J Am Coll Radiol* 2023;20:730–737 (doi 10.1016/j.jacr.2023.06.003)

- **Scope:** the ACR Pediatric AI Workgroup describes limited access to pediatric AI as a health equity issue.
- **Availability:** at the time of the review, only 6 of more than 200 FDA-cleared tools in the ACR catalog indicated pediatric use (about 3%).
- **Why adult AI may fail:** growth, disease patterns, and imaging protocols differ; published studies show uneven performance across pediatric subgroups.
- **Barriers:** scarce labeled pediatric data, consent and privacy concerns, and limited financial incentives in a smaller pediatric market.
- **Recommendations:** validate in children, disclose age-specific evidence, fund pediatric development, involve families, and account for pediatric workflows. The paper introduces the *Image IntelliGently* initiative.

Multisociety statement: safe pediatric AI implementation

Shelmerdine et al., *J Am Coll Radiol* 2026;23:89–101 (doi 10.1016/j.jacr.2025.08.019)

ACR, ESPR, SPR, SLARP, AOSPR, and SPIN make recommendations in four areas:

- **Regulation and purchasing:** review pediatric labeling, proposed safety ratings, training data, subgroup performance, transparency, and cybersecurity.
- **Implementation:** identify a clinical need, choose tools validated in children, run a local pilot, and involve pediatric specialists and families.
- **Monitoring after adoption:** keep clinicians responsible for interpretation, track performance and patient outcomes, review disagreements, report adverse events, and watch for overreliance on AI.
- **Education:** teach staff AI basics and pediatric applications, provide ongoing training, and involve patients and the public.

Why is pediatric radiology AI difficult?

- Small datasets
- Additional protections for children make clinical trials more complex
- Shortage of pediatric radiologists
- Lower expected financial return
- Adult-trained models may perform poorly in children
 - Different and more variable body sizes compared with adults
 - Different diseases
 - Different imaging protocols

Most radiology AI is developed for adults

Safe for kids? AI medical devices in radiology overlook paediatric suitability

Kelly et al., European Radiology 2026

Intended Population	Devices	Age (%)
Adult Only	79	37
Paediatric Only	4	2
All Ages	24	11
Subsets of Paediatric Patients	18	8
Unclear	88	41
Total	213	100

Intended population of all 213 regulator-approved radiology AI devices in the Health AI Register (European approvals). Only 4 (2%) are labeled for children only; 41% do not state a population, and after checking literature and marketing, 41 of those 88 were still unclear.

Performance of adult-trained AI models in paediatric imaging: a scoping review

Laborie et al., European Radiology 2026

- 20 studies (2014–2025) tested unmodified adult-trained models on pediatric images: radiography, CT, MRI, DEXA, ultrasound
- 18 of 20 found worse performance in children
- The largest declines were in detection: pulmonary nodule sensitivity fell from 68–100% in adults to 26–68% in children
- Segmentation: Dice scores fell by >0.10 across organs and modalities
- Performance declined most in children ≤ 2 years across all task types
- Test adult-trained tools in children before use; fine-tuning or retraining may be needed

What the ACR says

<https://www.acr.org/Data-Science-and-Informatics/Informatics/pediatric-radiology-ai-resources>

- Children have limited access to AI-assisted triage
- Pediatric payments through CHIP are historically 30% lower than adult Medicare, which may discourage pediatric AI development
- Adult-trained AI has shown mixed results in pediatric cases
- In 2023, only six of more than 200 FDA-cleared AI tools in the ACR Data Science Institute catalog (3%) were intended for pediatric use
- See the website for videos about reimbursement, challenges, and developments

What the ACR says

Take home points



Pediatric patients are left behind in radiology AI



There will not be equitable AI for all pediatric patients based on \$ ROI alone



Even in pediatrics, equitable is not one size fits all

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Resources for radiologists

- **Deep Learning: A Primer for Radiologists** (Radiographics paper, 2017): an introduction to how deep learning works, its radiology applications, and its limitations
- **RSNA Pediatric Bone Age Challenge (2017)** (benchmark / dataset): a public dataset and benchmark for testing automated bone-age assessment
- **Gleamer BoneView / AZmed Rayvolve** (clinical products): examples of fracture-detection assistance with pediatric indications; eligibility depends on age, anatomy, and product version
- **FDA AI-enabled device list and linked labeling** (regulatory resource): a starting point for checking authorized use and pediatric age ranges; the list is not exhaustive
- **Health AI Register** (product directory): commercial radiology AI products with CE / FDA status and intended use; confirm pediatric use in the labeling
- **CLAIM checklist (2024 update)** (reporting guideline): check whether data, reference standards, and testing are adequately described; reporting completeness alone does not establish clinical benefit
- **TotalSegmentator** (research tool): automated CT / MR organ segmentation for volumes and body composition; check accuracy in children before trusting measurements
- **nnU-Net** (research framework): a framework for training segmentation models and comparing new methods with an established baseline
- **MedSAM / segment-anything models** (research models): prompt-guided contours can assist annotation; radiologists still need to review and correct the output
- **MONAI** (research framework): software for developing and testing medical imaging AI with a research team
- **NVIDIA NV-Reason-CT** (open research model, Sept 2026): a 3D CT vision-language model that drafts chest / abdominal CT findings with step-by-step reasoning and answers questions; open weights, research use only, not FDA-cleared, and no pediatric data or age range reported

Explore our database: <https://pedrad-ai.vercel.app>

Pediatric Radiology AI Database

Articles, open source software, trade-press stories, commercial devices and public datasets for AI in pediatric radiology. Search, filter columns by value, sort, pick columns, and export any view as CSV.

Articles 3,496Open source software 339Newsletters 161Commercial 997Datasets 13

The FDA's list of AI-enabled medical devices, Radiology panel only (snapshot 2026-09-17; [source](#)), one row per device: a device the FDA authorized more than once (a new version or indication) is shown at its most recent authorization, with the earlier ones listed, so the 1,230 authorizations make 994 rows. The FDA describes the list as noncomprehensive. The 3 rows marked **not FDA-authorized** are pediatric products on our curated list that have no US authorization (for example BoneXpert, CE-marked and US research use only). **Pediatric use** is yes for the 172 authorized devices with at least one authorization whose linked label states a pediatric or fetal patient population or intended use (2026-09-16 screen); hover the yes for which. About half are whole scanners whose labels list pediatric imaging as one clinical application, so confirm the current authorization before purchase. **Curated pediatric product** marks the 23 products with a documented pediatric indication reviewed by hand on 2026-09-16; on an FDA row it is a company-level flag, not a claim about that row's authorization.

SearchClearColumnsExport CSV

Examples: `pediatric:yes` `company:="GE HealthCare"` `date>=2025` `fetal OR pediatric` `us_status:="not FDA-authorized"` [show fields](#)

12 of 997 rows match.

Device	Company	Decision date	Submission	US status	Earlier authorizations	Product code	Pediatric-use screen	Pediatric use
▶ EFAI Chestsuite XR Malpositioned ETT Assessment System (ETT-XR-100)	Ever FortuneAI	2025-02-20	K242821	FDA-authorized	—	QAS	label-positive candidate	yes
▶ EFAI Bonesuite XR Bone Age Pro Assessment System (BAP-XR-100)	Ever FortuneAI	2024-06-07	K234042	FDA-authorized	—	QIH	label-positive candidate	yes
▶ EFAI CARDIOSUITE CTA ACUTE AORTIC SYNDROME ASSESSMENT SYSTEM	Ever FortuneAI	2024-04-08	K240291	FDA-authorized	—	QAS	label-positive candidate	yes
▶ EFAI ERSUITE CT APPENDICITIS ASSESSMENT SYSTEM (APPEN-CT-100)	Ever FortuneAI	2026-04-22	K253163	FDA-authorized	—	QAS	—	no
▶ Seg Pro V3 (RT-300)	Ever FortuneAI	2026-01-28	K251306	FDA-authorized	—	QKB	—	no
▶ EFAI Neurosuite CT Midline Shift Assessment System (MLS-CT-100)	Ever FortuneAI	2024-12-06	K241923	FDA-authorized	—	QAS	—	no
▶ EFAI NeuroSuite CT ICH Assessment System	Ever FortuneAI	2023-10-04	K231025	FDA-authorized	—	QAS	—	no
▶ EFAI RTSUITE CT HCAP-Segmentation System	Ever FortuneAI	2023-09-25	K231928	FDA-authorized	—	QKB	—	no
▶ EFAI ChestSuite XR Pneumothorax Assessment	Ever FortuneAI	2022-11-08	K221552	FDA-authorized	—	QFM	—	no

Recommendations for pediatric radiologists

- Pediatric AI studies often have few cases, especially for rare diseases, with data spread across hospitals and inconsistent reporting.
- Start with a specific task in your practice: bone-age measurements, fracture detection, or reducing CT dose / MRI scan time.
- Before adopting a tool, ask which children it was tested on. Check the intended age range and test it on your own cases, including younger children and unusual anatomy.
- In a local pilot, measure what changes with AI: reading time, missed findings, false alarms, dose, or scan time. Review errors and keep checking after rollout.
- For research, choose a pediatric problem you see often, label cases carefully, and work with other hospitals to test whether a model works beyond the site that built it.